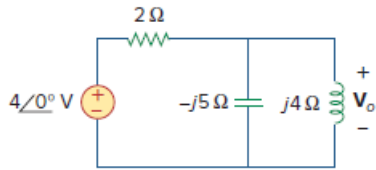


Problem

Using Fig. 10.51, design a problem to help other students better understand nodal analysis.

Figure 10.51:



Step-by-step solution

Step 1 of 2

Refer to Figure 10.51 in the textbook.

Consider nodal analysis is applied with a combination of ohms law and Kirchhoff's current law.

To apply nodal analysis in the circuit, consider a node in the circuit and then apply Kirchhoff's current law at that node.

From the Figure 10.51 consider a node V_o in the circuit as shown in Figure 1.

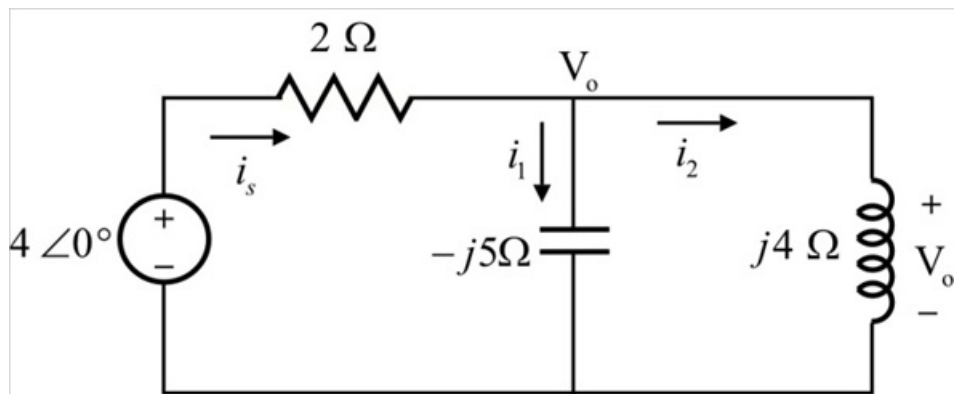


Figure 1

Step 2 of 2

Apply Kirchhoff's current law at node V_o in the Figure 1.

Consider i_s is the current entering at node V_o and i_1, i_2 are current leaving from node V_o .

$$i_s = i_1 + i_2$$

$$\frac{V_o - 4\angle 0^\circ}{2} + \frac{V_o}{-j5} + \frac{V_o}{j4} = 0$$

$$V_o \left[\frac{1}{2} - \frac{1}{j5} + \frac{1}{j4} \right] = \frac{4\angle 0^\circ}{2}$$

$$V_o [0.5 + j0.2 - j0.25] = 2$$

$$V_o [0.5 - j0.05] = 2$$

$$V_o = \frac{2}{0.5 - j0.05}$$

$$= \frac{2}{0.5\angle -5.71^\circ}$$

$$= 4\angle 5.71^\circ$$

Therefore, the node voltage V_o is $4\angle 5.71^\circ \text{ V}$.